MIT | **Academy of Engineering**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Subject (Code):
Essentials of Data Science
(2304102L)

EDS CodeTantra Programs

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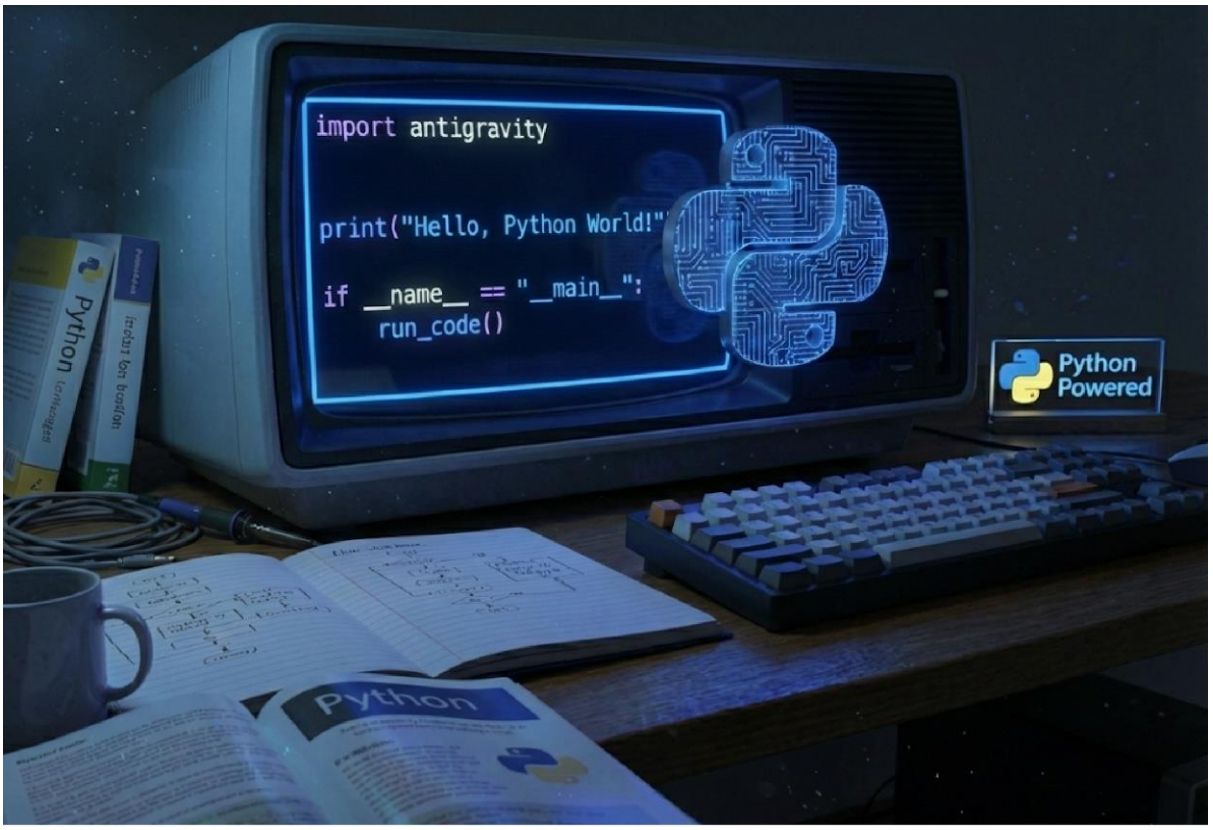


Practical 05

5.1 Practice Lab Assignments

5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

```
import matplotlib.pyplot as plt
import pandas as pd

# Data for Months and Temperature for three cities
data = {
    'Month': ['January', 'February', 'March', 'April', 'May', 'June',
'July', 'August', 'September', 'October', 'November', 'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}

# Create a DataFrame
df = pd.DataFrame(data)

# Create the stacked area plot
plt.stackplot(df['Month'], df['City_A_Temperature'],
df['City_B_Temperature'], df['City_C_Temperature'])

# Add labels and title
plt.xlabel('Month')
plt.ylabel('Temperature')
plt.title('Temperature Variation')

# Display the plot
plt.show()
```

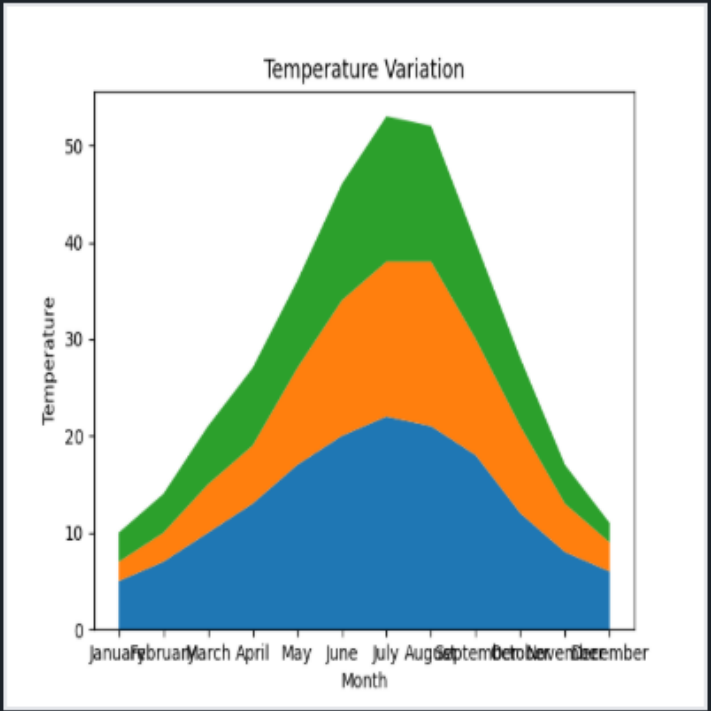
Average time
0.178 s
178.00 ms

Maximum time
0.178 s
178.00 ms

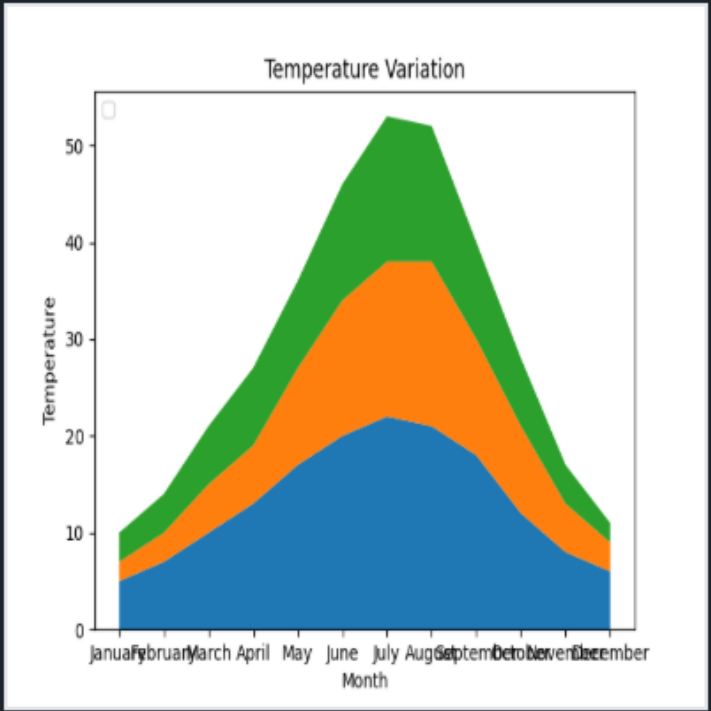
1 out of 1 shown test case(s) passed

Test case 1 178 ms DEBUG

Expected output



Actual output



5.2 Lab Assignments

5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:
The dataset is stored in a CSV file named titanic.csv and has been loaded using the pandas library. It contains the following columns:

- Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).
- Gender: Gender of the passenger (male/female).
- Age: Age of the passenger.
- Survived: Survival status (0 = Did not survive, 1 = Survived).
- Fare: Ticket fare paid by the passenger.

Visualization:
To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:
Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):
- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).
- Use the color skyblue for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution):
- Create a pie chart to display the distribution of male and female passengers.
- Use lightblue for males and lightcoral for females.
- Include percentages on the slices (use autopct='%1.1f%%').
- Title the plot as **"Gender Distribution"**.

Histogram (Age Distribution):
- Create a histogram to visualize the distribution of passengers' ages.
- Use lightgreen for the bars with black edges (edgecolor = 'black').
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"**.
- Label the x-axis as **"Age"** and the y-axis as **"Frequency"**.

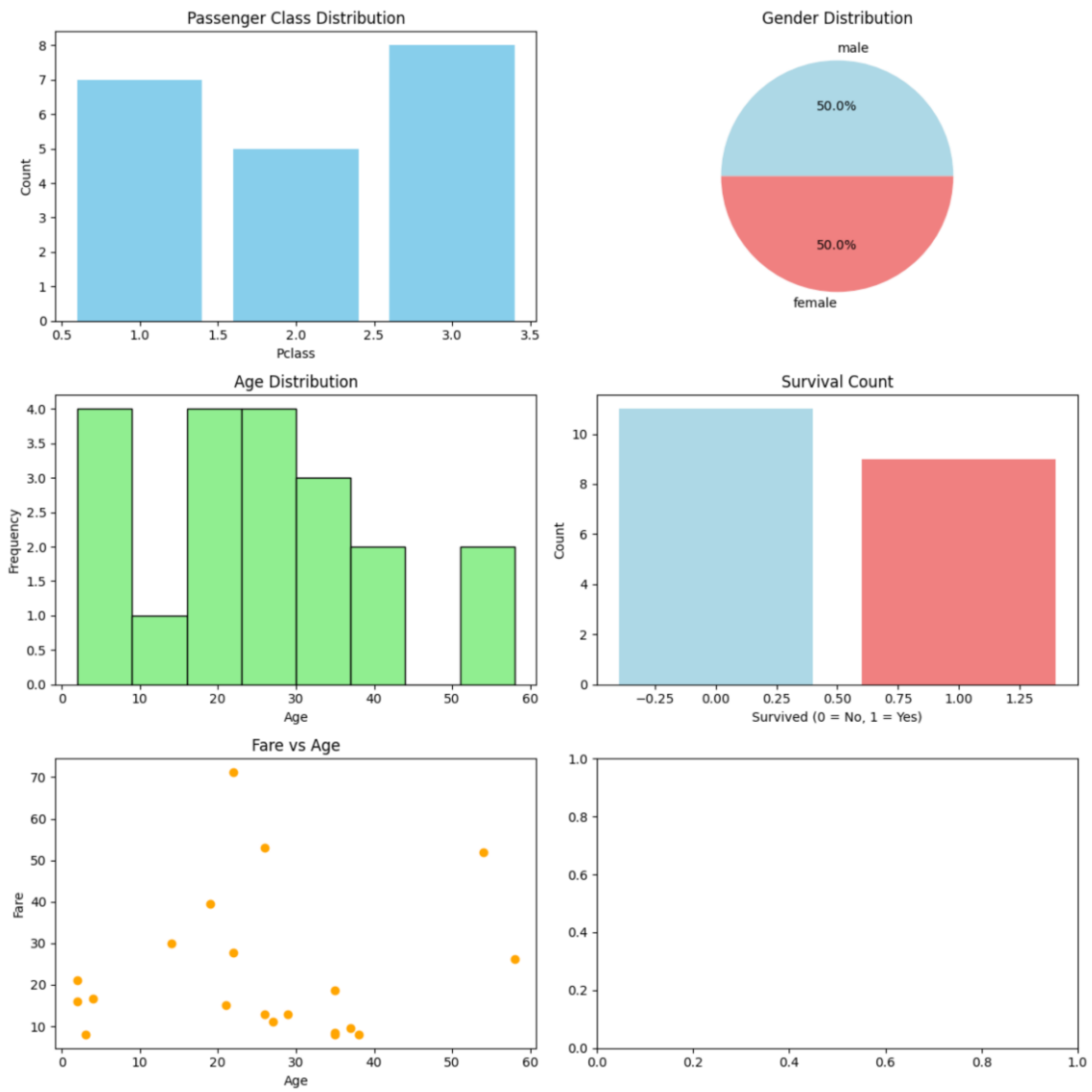
Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column.
- Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0).
- Title the plot as **"Survival Count"**.
- Label the x-axis as **"Survived (0 = No, 1 = Yes)"** and the y-axis as **"Count"**.

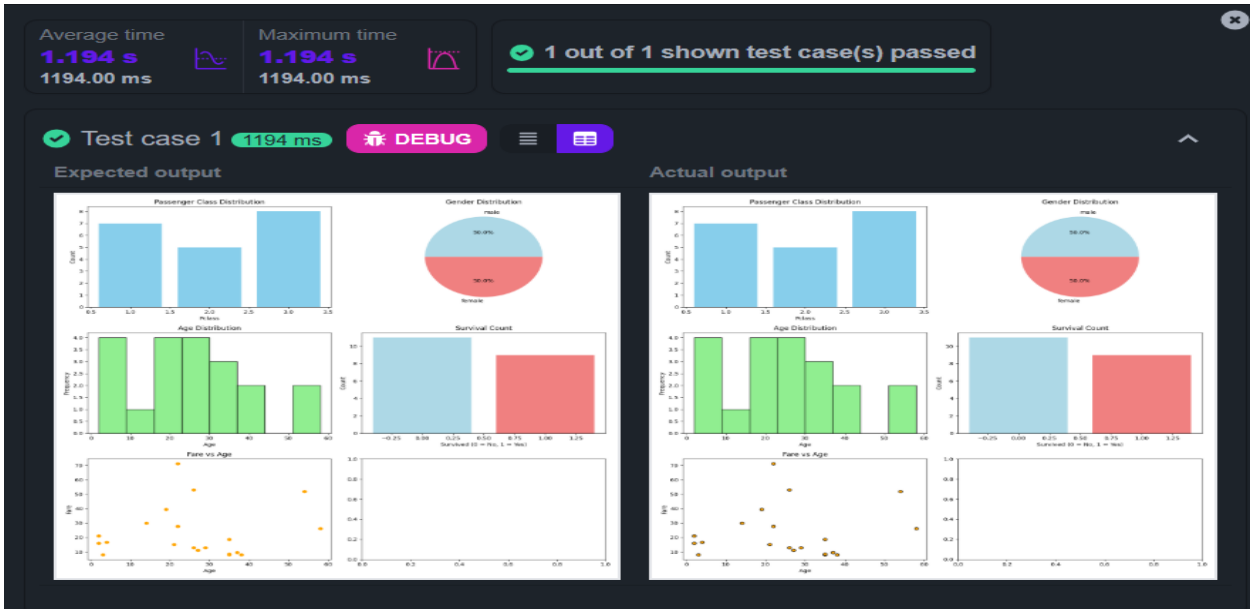
Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as **"Fare vs Age"**.
- Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

Note: Refer to the displayed plot in the sample test cases for better understanding.



```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# write the code..

axes[0,0].bar(df['Pclass'].value_counts().index,df['Pclass'].value_counts(), color='skyblue')

axes[0,0].set_title("Passenger Class Distribution")

axes[0,0].set_xlabel("Pclass")

axes[0,0].set_ylabel("Count")


axes[0,1].pie(df['Gender'].value_counts(),labels=df['Gender'].value_counts().index, autopct='%1.1f%%',colors=['lightblue','lightcoral'])

axes[0,1].set_title("Gender Distribution")


axes[1,0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')

axes[1,0].set_title("Age Distribution")

axes[1,0].set_xlabel("Age")

axes[1,0].set_ylabel("Frequency")


axes[1,1].bar(df['Survived'].value_counts().index,df['Survived'].value_counts(),color=['lightblue','lightcoral'])

axes[1,1].set_title("Survival Count")

axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")

axes[1,1].set_ylabel("Count")
```



5.2.2. Histogram of Passenger Information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black** (k).
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

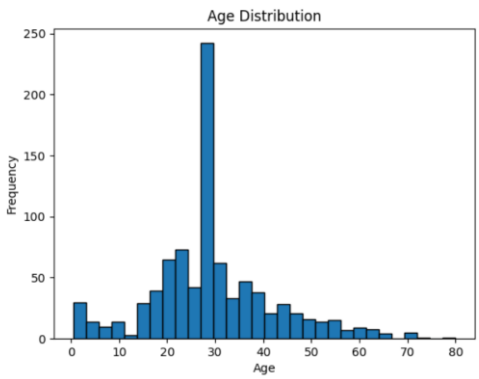
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2.
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)" female, 35, 1, 0, 113803, 53.
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt
```

```python
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the Histogram
plt.hist(data["Age"].dropna(), bins=30, edgecolor= "k")

# Add labels and title
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution")

# Display the plot
plt.show()
```

5.2.3. Bar Plot of Survival rate of Passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

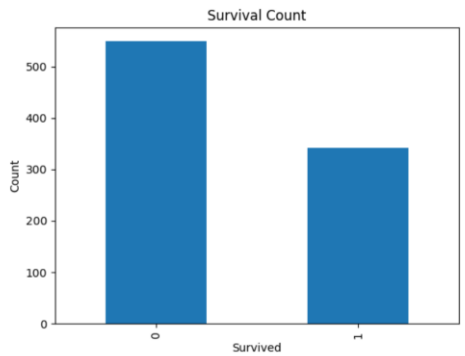
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

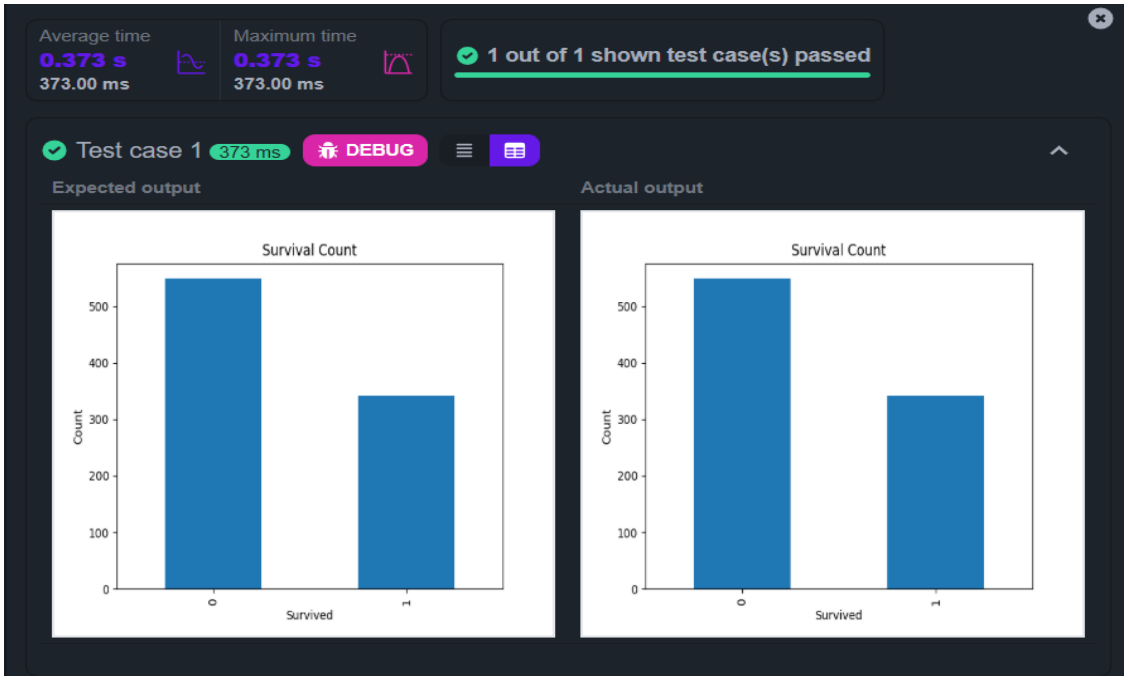
Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.



```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Count the number of survivors and non-survivors
survival_counts = data['Survived'].value_counts().sort_index()

# Create the bar plot
survival_counts.plot(kind='bar')

# Add title and labels
plt.title("Survival Count")
plt.xlabel("Survived")
plt.ylabel("Count")

# Display the plot
plt.show()
```



5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend The Titanic should indicate **'Not Survived'** and **'Survived'**.

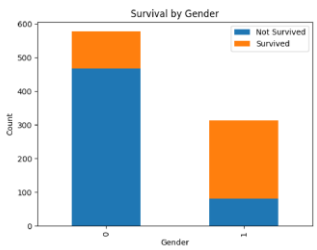
dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

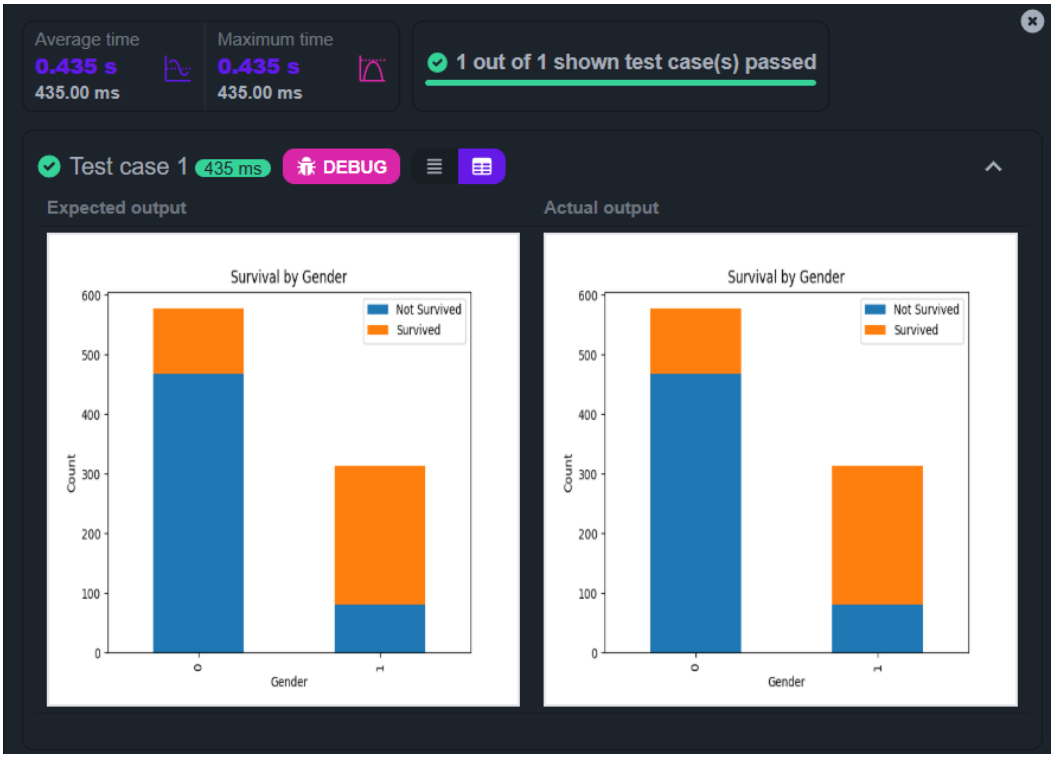
Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.



```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning

data['Age'].fillna data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Prepare the data. Group by Sex and Survived, then count the size of
each group and unstack to get columns for 0 and 1
survival_gender = data.groupby(['Sex', 'Survived']).size().unstack()

# Create the stacked bar chart
ax = survival_gender.plot(kind='bar', stacked=True)

# Add title and labels
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Count")

# Customize Legend
plt.legend(["Not Survived", "Survived"])

# Display the plot
plt.show()
```



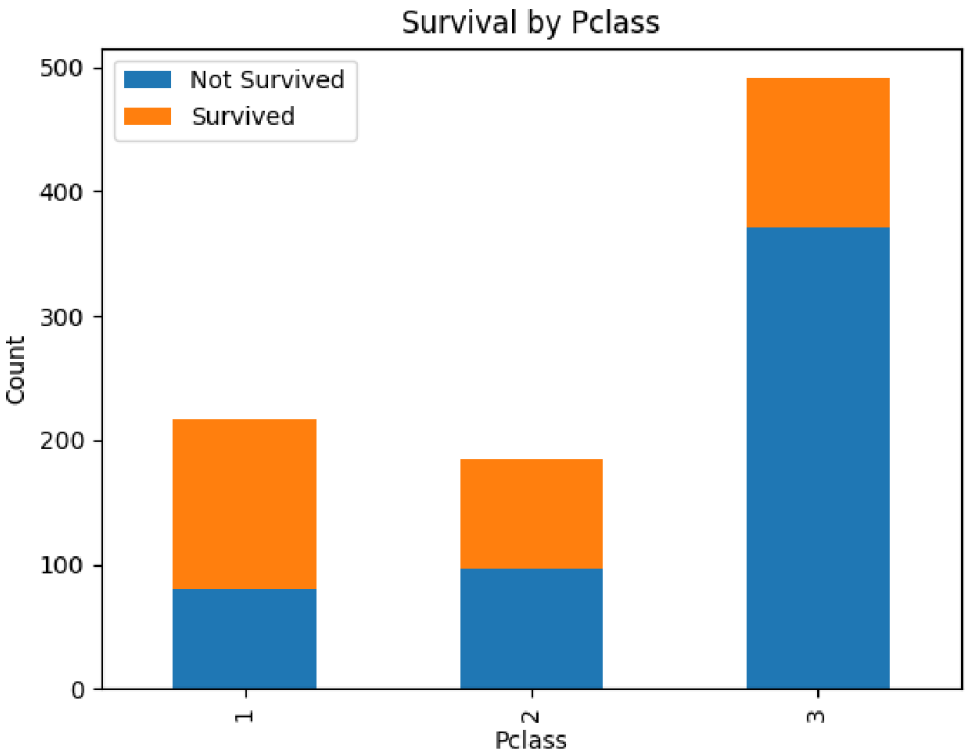
5.2.5. Bar Plot for Survival by Class

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked



Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

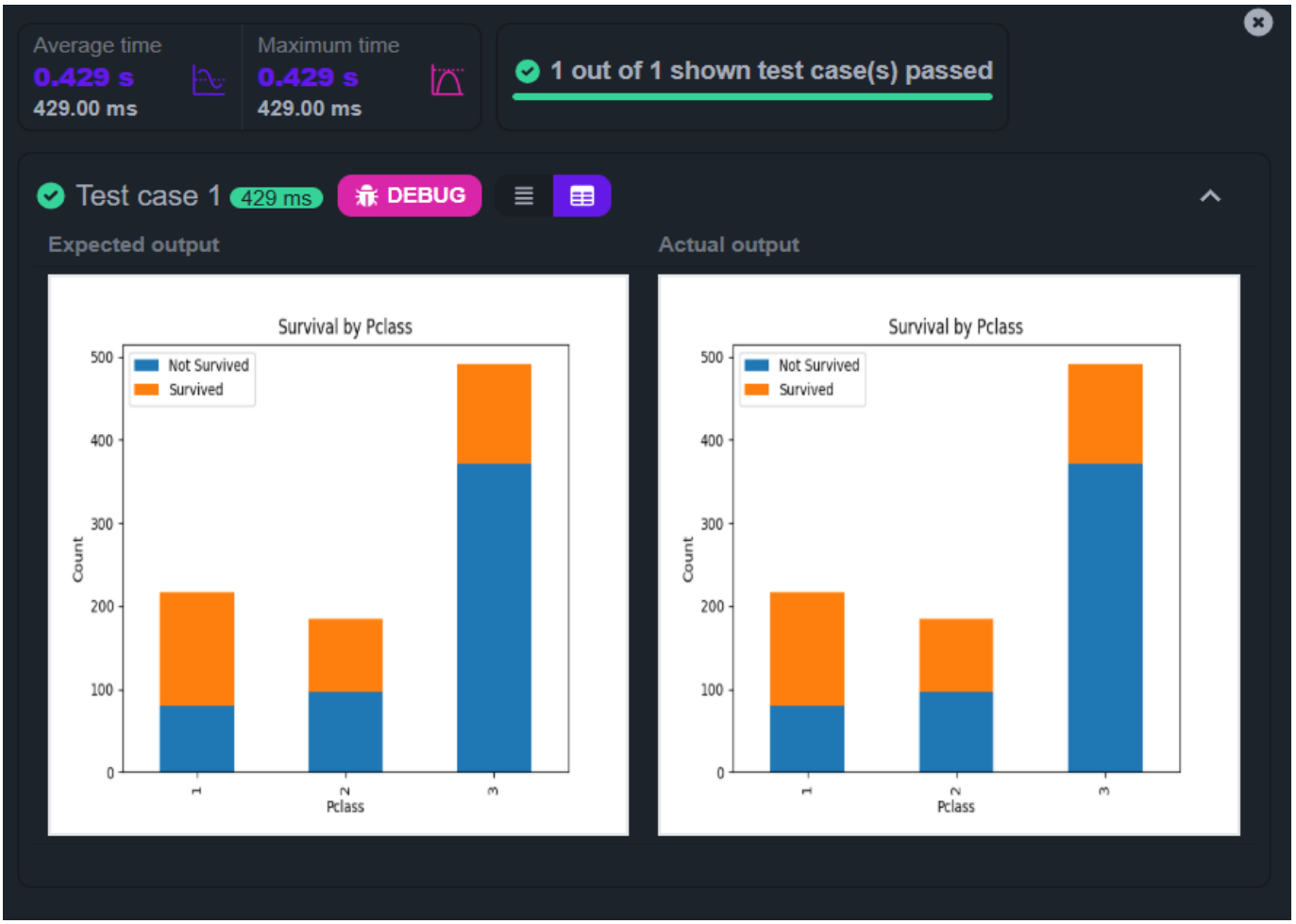
Note:

- Refer to the visible test case for better reference.
- Ensure you use the **groupby()** function with **value_counts()** to count the survivors and non-survivors for each **Pclass**.
- Do **not** manually use **size()** or **unstack()** without **value_counts()**. Use the **value_counts()** method for counting survival status directly.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# Write your code here for Bar Plot for Survival by Pclass

survival_counts=data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Pclass')

plt.xlabel('Pclass')

plt.ylabel('Count')

plt.legend(['Not Survived','Survived'])

plt.show()
```



5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked "** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

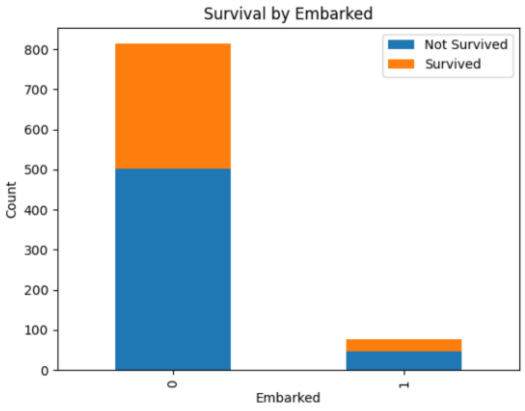
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

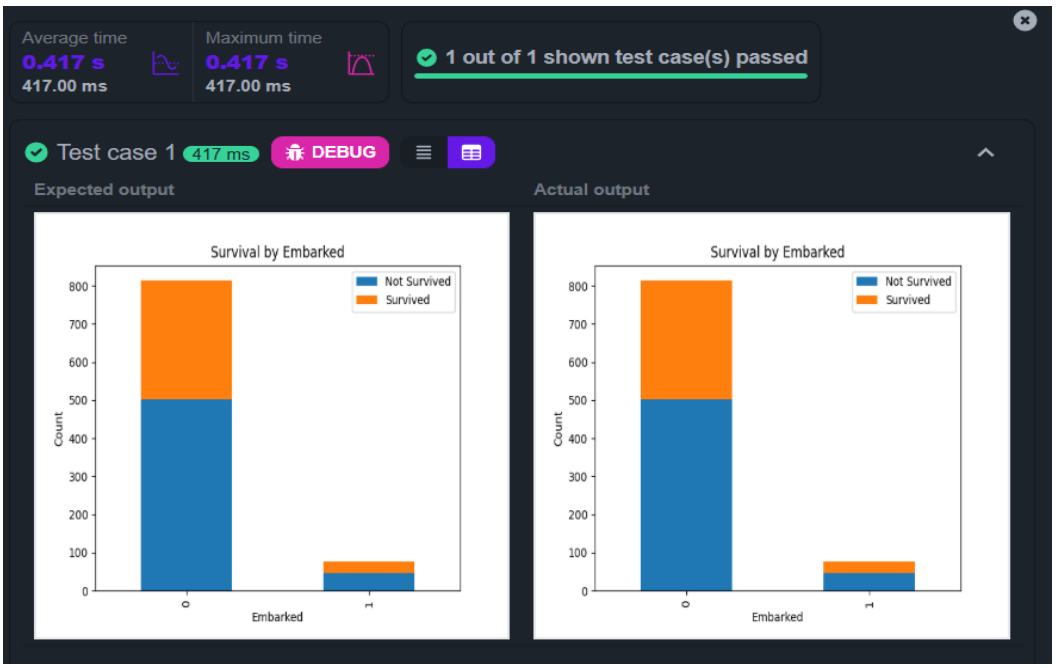
Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W(Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age']. fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

survival_counts = data.groupby('Embarked_Q')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Embarked')

plt.xlabel('Embarked')

plt.ylabel('Count')

plt.legend(['Not Survived','Survived'])

plt.show()
```



5.2.7. Box Plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

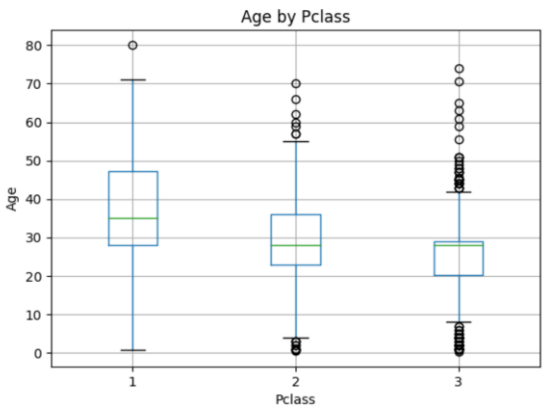
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W(Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
data.boxplot(column='Age', by='Pclass')
plt.title('Age by Pclass')
plt.suptitle('')
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.show()
```

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

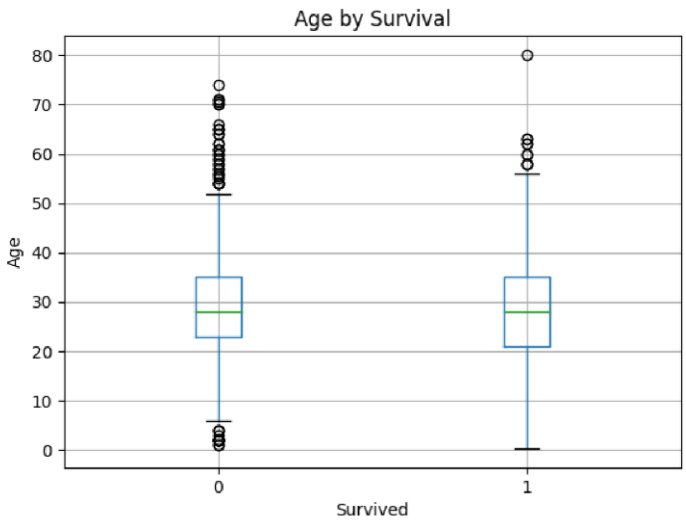
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

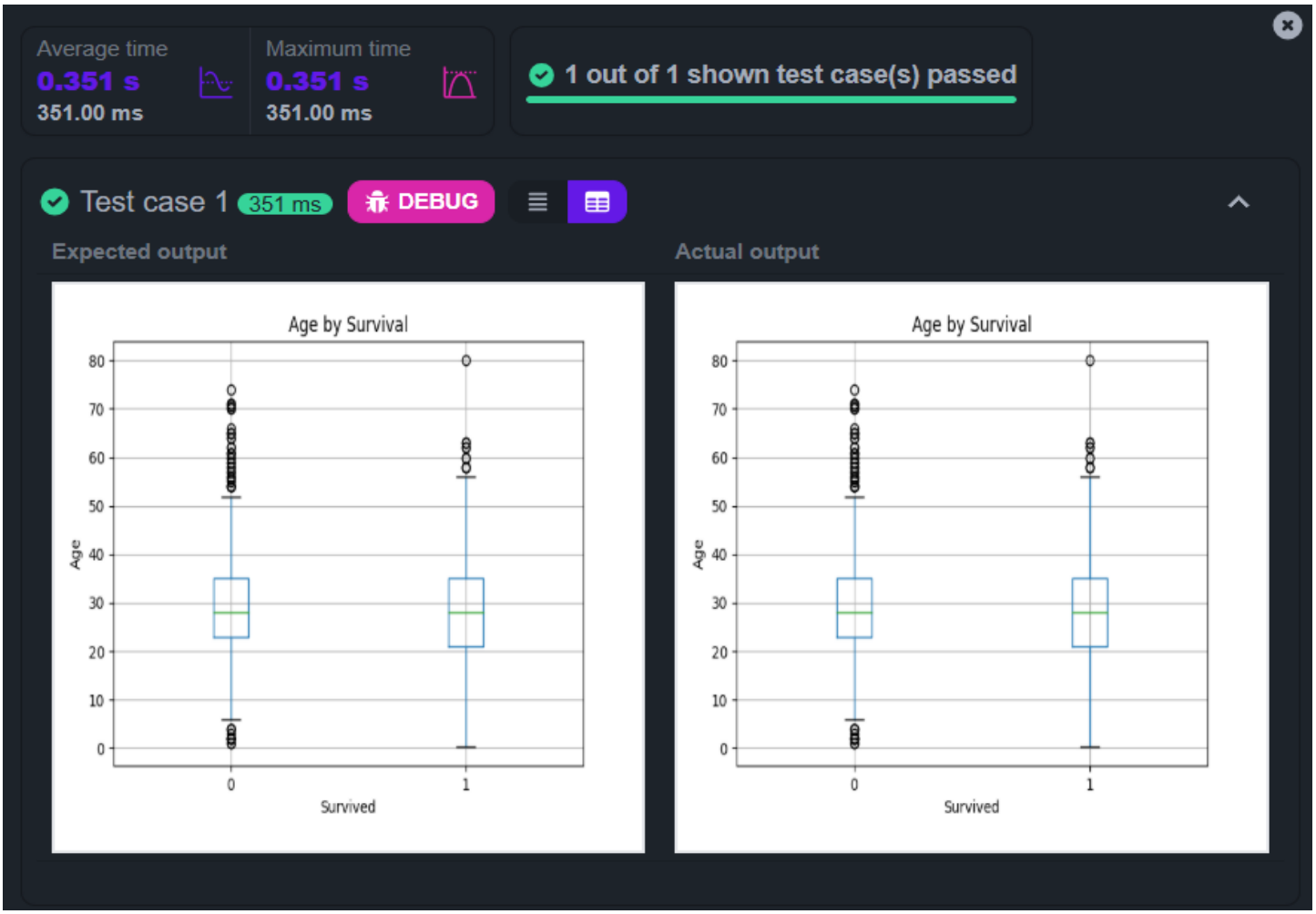
Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the boxplot for Age by Survived
data.boxplot(column='Age', by='Survived')

# Customize titles and labels
plt.title("Age by Survival")
plt.suptitle('')  # Removes the default "Boxplot grouped by Survived"
subtitle
plt.xlabel("Survived")
plt.ylabel("Age")

# Display the plot
plt.show()
```



5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

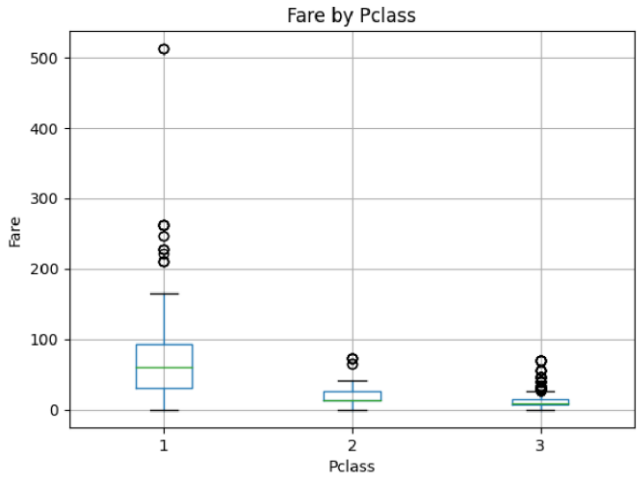
PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs

Thayer)", female, 38, 1, 0, PC 17599, 71. 2833, C85, C

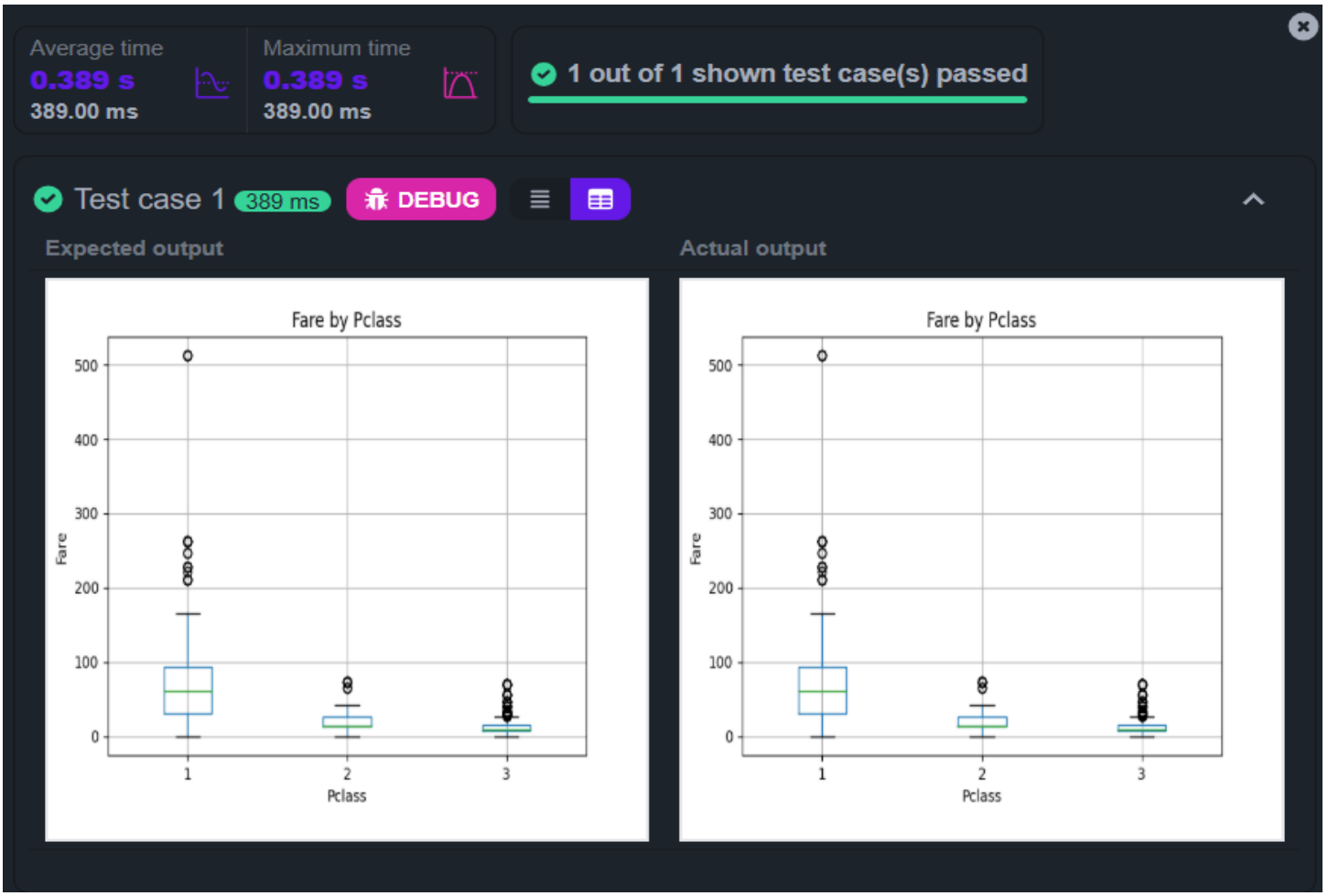
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/ O2. 3101282, 7. 925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53. 1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8. 05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8. 4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51. 8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21. 075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W(Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11. 1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30. 0708, , C



Note: Refer to the visible test case for better reference

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the boxplot for Fare by Pclass
data.boxplot(column='Fare', by='Pclass')

# Customize titles and labels
plt.title("Fare by Pclass")
plt.suptitle('')  # Removes the default "Boxplot grouped by Pclass"
subtitle
plt.xlabel("Pclass")
plt.ylabel("Fare")

# Display the plot
plt.show()
```



5.2.10. Scatter Plot for Age vs Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

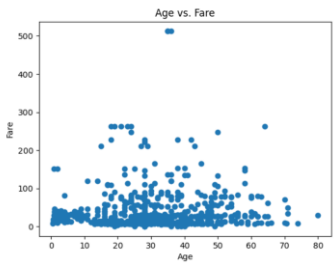
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.



```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning



data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the scatter plot for Age vs Fare
plt.scatter(data['Age'], data['Fare'])

# Add title and labels
plt.title("Age vs. Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

# Display the plot
plt.show()
```

5.2.11. Scatter Plot for Age vs Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

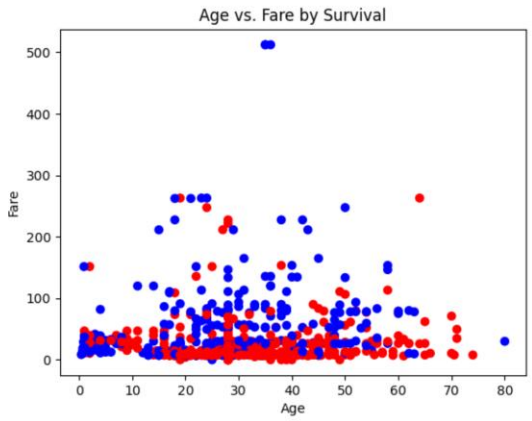
The Titanic dataset contains columns as shown below,

PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

Sample Data:

- PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- 1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
- 2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
- 3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
- 4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
- 5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
- 6, 0, 3, "Moran, Mr. James", male, , 0, 0, 330877, 8.4583, , Q
- 7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
- 8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, , S
- 9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, , S
- 10, 1, 2, "Nasser, Mrs. Nicholas (Adele Achem)", female, 14, 1, 0, 237736, 30.0708, , C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)



# Create a color



colors = data['Survived'].map({0: 'red', 1: 'blue'})

# Create the scatter plot for Age vs Fare using the c argument for
colors
plt.scatter(data['Age'], data['Fare'], c=colors)

# Add title and labels
plt.title("Age vs. Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")

# Display the plot
plt.show()
```

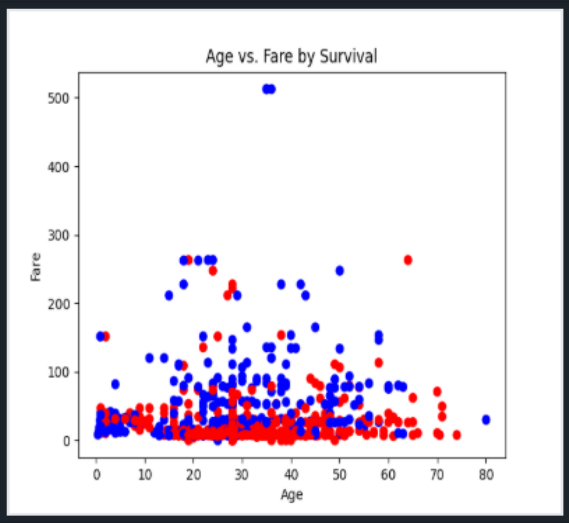
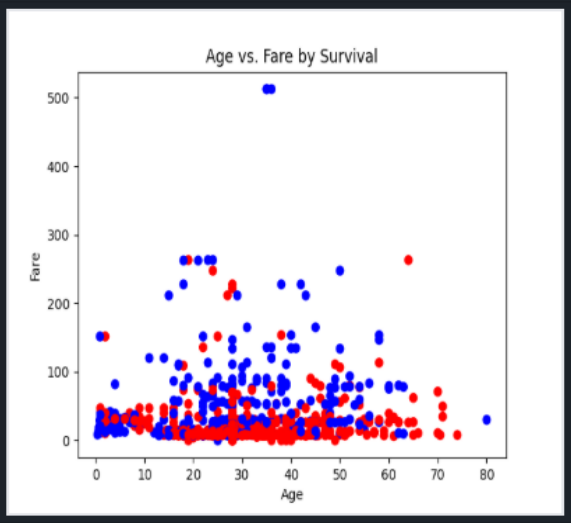
Average time
0.404 s
404.00 ms

Maximum time
0.404 s
404.00 ms

1 out of 1 shown test case(s) passed

Test case 1 404 ms DEBUG

Expected output



Actual output